Internship Report

During my internship, my primary focus was to extend my postgraduate dissertation project by applying transfer learning to my pothole detection model using YOLOv8 (the best model from my dissertation), particularly by using the UK dataset.

The aims and objectives of this project were:

- Experiment 1:
  Use the model trained with the US dataset and test it with the 21 UK images (completed).

- Experiment 2:
  Use my dissertation model, YOLOv8n (with double augmentation), and apply transfer learning by training on 20 images and testing the trained model on the remaining image. Repeat this process 21 times, each time testing on a different image.

  Model 1: Transfer learning using a pre-trained YOLOv8n (on the US dataset) without augmentation.
  Model 2: Transfer learning using a pre-trained YOLOv8n (on the US dataset) with YOLO-integrated augmentations.
  Model 3: Transfer learning using a pre-trained YOLOv8n (on the US dataset) with Roboflow augmentations.
  Model 4: Transfer learning using a pre-trained YOLOv8n (on the US dataset) with both YOLO integrated and Roboflow augmentations.

Initially, the experiment went smoothly by testing the model trained on the US dataset with the 21 available images from the UK dataset. After running this experiment, I proceeded to implement transfer learning using these UK images, iterating with different augmentation strategies.

Model 1, applying transfer learning without augmentation, wasn't much of a hurdle. However, issues arose when I had to apply augmentation. Due to the limited data available to train YOLO and produce a stable model that could be tested on a sufficient number of images, as well as the need to compare all four models and select the best one, it became challenging to execute this experiment given the time frame and resources.

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In the end, I completed two models:

Model 1: Transfer learning applied to the pre-trained model from my postgraduate dissertation without augmentation.
Model 2: Transfer learning applied to the pre-trained model from my postgraduate dissertation with YOLO integrated augmentation.

Throughout the internship, I encountered challenges such as managing a small dataset. While many techniques exist to balance and increase the dataset size, such as augmentation and synthetic data generation, the limited resources (e.g., free Google Colab access with restricted GPU usage) and time constraints hindered my ability to implement these methods fully. By using transfer learning, applying augmentation, carefully managing the dataset splits and performing training in chunks, I was able to improve the model's performance. However, further work is needed to fine-tune the model and address overfitting.

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